Impact & Benefits

07

Dual-Dimension Evaluation: Multi-Stakeholder Operational Impact & Environmental, Social, and Governance (ESG) Framework Based Benefits

MULTI-STAKEHOLDER OPERATIONAL IMPACT

Targeted operational enhancements delivered across critical disaster relief, command, and community entities



Tactical Rescue Squads — NDRF / USAR

- **Hazard-Aware GPS Routes:** Steers inflatable boats & amphibious craft away from fast-moving flash currents and submerged debris.
- **Building Entrapment Triage:** Replaces blind sweeps with deterministic building risk scores (Ri) to rescue trapped victims first.

Disaster Command Center — NDMA / SDMA

- **Unified GIS Dashboard:** Fuses cloud-penetrating SAR flood extent with optical building footprints in a single live operational pane.
- **Automated Squad Tasking:** Generates auditable digital rescue tickets with verifiable satellite metrics for multi-agency sync.

Stranded Communities — LOCAL CITIZENS

- **Rapid Enclave Localization:** Discovers isolated dwellings obscured by dense cloud cover or night storms within hours.
- **Equitable Life Triage:** Impartial algorithmic scoring guarantees remote rural and informal settlements receive equal priority.

Emergency Medical (EMS) — FIELD HOSPITALS

- **Predictive Casualty Influx:** Estimates emergency surgical and trauma loads directly from structural damage levels (Si).
- **Clinic Inundation Alerts:** Monitors rising flood corridors threatening primary healthcare centers, oxygen units, and power grids.

Municipal Infrastructure — UTILITIES / PWD

- **Critical Lifeline Monitoring:** Detects breached levees, inundated electrical substations, washed causeways, and blocked arterials.
- **Secondary Collapse Warning:** Flags compromised structures vulnerable to waterlogged foundation failure and imminent collapse.

Humanitarian Relief NGOs — RED CROSS / OCHA

- **Targeted Supply Routing:** Directs drinking water, food packets, and medical kits straight to verified stranded clusters.
- **Transparent Relief Accounting:** Prevents aid hoarding and misallocation through tamper-resistant geospatial delivery logs.

ESG BENEFITS

Alignment with Environmental, Social, and Governance global sustainability criteria

E — ENVIRONMENT

Hydrological Dynamics & Eco-Protection

- **Dynamic Hydrological Mapping:** Delineates standing water vs newly intruded flood expansion across sensitive wetland basins.
- **Carbon-Optimized Deployment:** Replaces blind search sweeps with direct GPS triage, cutting unnecessary boat & helicopter fuel burn.

S — SOCIAL

Humanitarian Equity & Life Preservation

- **Unbiased Life-Saving Priority:** Objective algorithmic risk scoring eliminates socioeconomic and demographic favoritism in aid allocation.
- **First-Responder Hazard Safety:** Warns rescue crews of severely collapsed structures (Class 3) before squads enter dangerous collapse zones.

G — GOVERNANCE

Transparent Explainability & Public Trust

- **100% Explainable AI (Grad-CAM):** Saliency heatmaps demystify deep learning inference for civilian commanders and oversight boards.
- **Auditable Mathematical Standards:** Deterministic formula (Ri = 0.5Ei + 0.5Si) and timestamped GIS incident logs ensure public trust.